

Phase I: Controlled English, Hindi, and Hinglish Benchmark

Research findings report | Provisional pending translation review | Generated from the completed 216-cell study

216	27	3	0
Gradable cells	Strict successes	Instruction languages	Infrastructure gaps

Executive finding

The university-GPU ReAct agent achieved the same strict task success rate in all three instruction languages: 9 of 72 cells per language, or 12.5%. The observed language deltas are therefore zero in this study. The result is best interpreted as evidence of no measured language effect under this fixed model, agent, task sample, and translation version - not as evidence that the languages are universally equivalent.

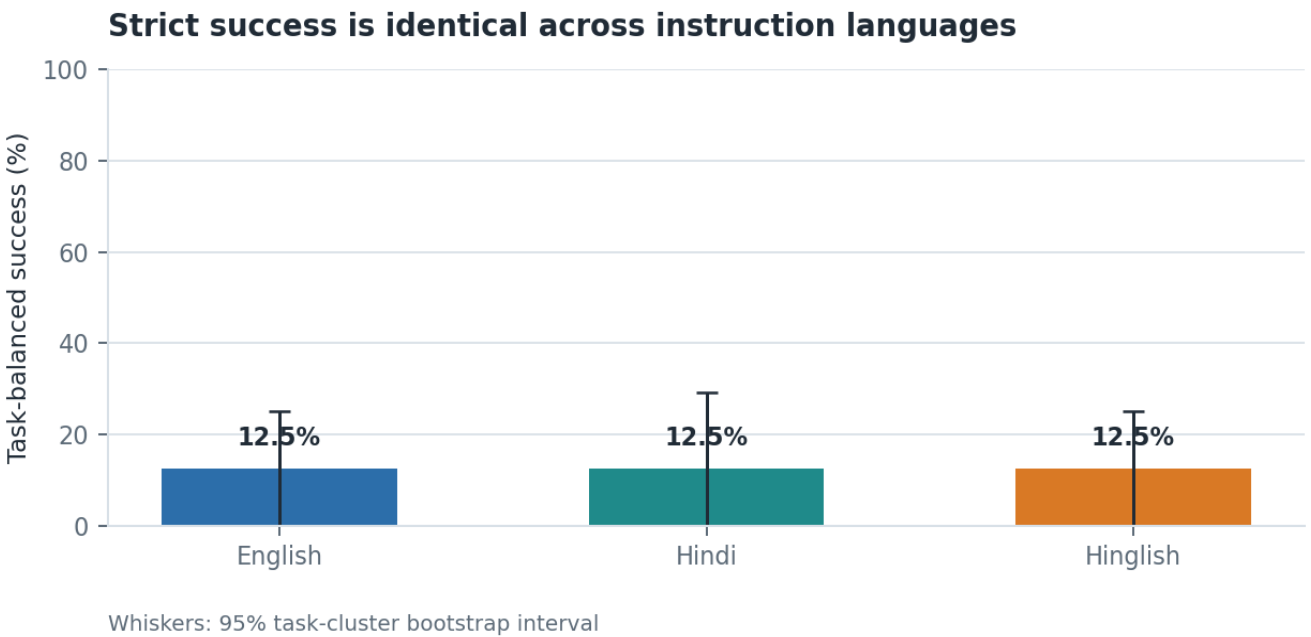


Figure 1. Strict success by language. Bars show task-balanced success; whiskers show 95% task-cluster bootstrap intervals.

How to read this report

A cell is one task-language-repetition condition. Success uses each upstream task’s strict expected outcome threshold or judge exit code. Infrastructure failures are excluded from success rates; the final database contains none. Hindi and Latin-script Hinglish overlays were authored before execution but are still marked unreviewed, so all findings are provisional.

1. Study design and provenance

Item	Fixed value
Benchmark source	Harness-Bench pinned at commit 1025086a446653702b80cfb48babbeec35db6b2c; 24 local task definitions and source hashes.
Matrix	24 tasks x 3 languages x 3 repetitions = 216 stable cells; shuffle seed 17; sequential execution.
Languages	English, Hindi, and natural Latin-script Hinglish. English prompts were preserved verbatim.
Agent and generation	One ReAct agent; temperature 0; top-p 1; maximum 2,048 generated tokens per model call; maximum 40 agent steps; task-specific timeouts.
Inference	University GPU OpenAI-compatible endpoint. The served model identity and tool-call check were pinned locally; the private endpoint, key, and server-backed model identifier are intentionally omitted from this report.
Isolation	Fresh fixtures per cell; agent shell commands in a network-disabled Docker container; final workspace graded in a separate container with the oracle unavailable to the agent.
Image	indic-harness-phase1:2026-09-21; digest 21ecab43c35b707512e8ad137343fb1fbea45d1059c159d26b6947affe8f74af.

Preflight and pilot

The five-task pilot covered 15 runs across the same three languages and passed infrastructure checks. It identified and corrected two runner issues before the main study: provider-side malformed tool-call JSON was recorded as a model/tool error, and large or exception-bearing oracle output was preserved as a gradable score rather than lost as infrastructure. The main study was then resumed from stable cell identities without duplicating completed cells.

Completion audit

Audit item	Observed	Interpretation
Stable database cells	216 completed	Meets the Phase I completion criterion.
Grades	216	Every completed cell has a grade record.
Trace and result files	216 JSONL traces + 216 result JSON files	One per stable cell; indexed locally by cell_id.
Event records	9,381	Tool arguments/results and model-call metadata retained without private chain-of-thought.
Failed tool results	1,078 across retained attempts	Includes command exit failures; final-cell counters exclude superseded retry attempts.
Secret scan	No private marker in tracked files or experiment traces	Private runtime values remain in ignored local storage.

2. Main outcomes

Language	Successful cells	Gradable cells	Task-balanced rate	95% task-cluster interval
English	9	72	12.5%	[0.0%, 25.0%]
Hindi	9	72	12.5%	[0.0%, 29.2%]
Hinglish	9	72	12.5%	[0.0%, 25.0%]

Paired outcomes

Across the 72 task-repetition groups, 9 were all-pass across all three languages and 63 were not all-pass. The current binary outcome pattern contains no language-specific divergence: the same three tasks passed in every language and every repetition.

Paired outcome	Groups
all pass	9
mixed	63

The three uniformly successful tasks were 001-file, 005-email-triage, and 020-archive-checksum. All other selected tasks failed the strict success threshold in all three language conditions, although many received partial oracle scores.

3. Efficiency and interaction metrics

Operational cost and interaction profile

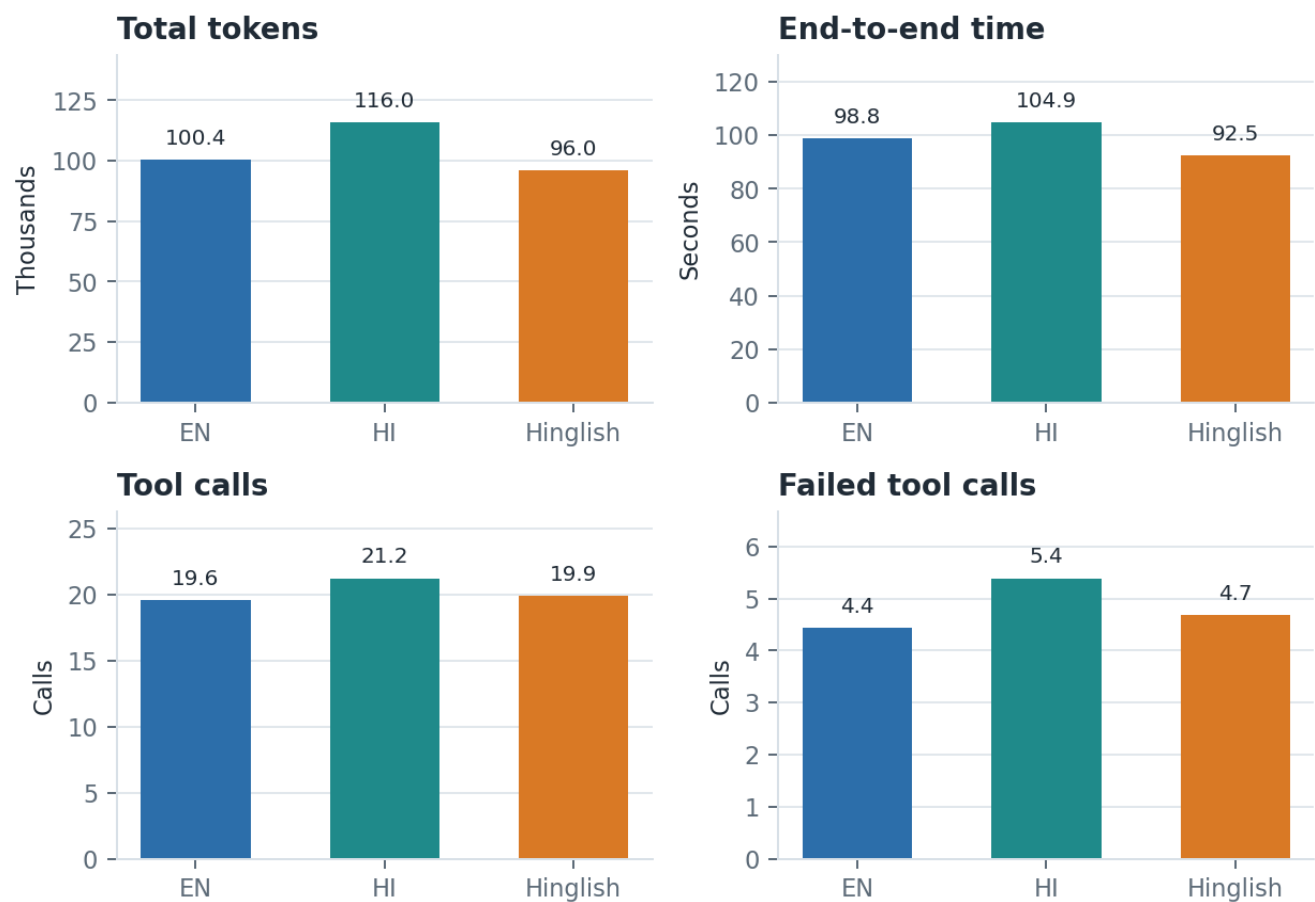


Figure 2. Operational profile. Bars show per-cell means. These are descriptive comparisons, not independent significance tests.

Metric	English	Hindi	Hinglish
Initial-prompt tokens	1,071.5	1,275.3	1,072.1
Total tokens	100,374.9	116,004.0	95,992.5
Agent time (s)	96.5	102.6	90.2
End-to-end time (s)	98.8	104.9	92.5
Tool calls	19.6	21.2	19.9
Failed tool calls	4.4	5.4	4.7
Recovery after error	91.7%	95.8%	91.7%

Interpretation

Hindi used more initial-prompt and total tokens on average than English, while Hinglish used fewer total tokens and slightly less time. Hindi also had the highest average failed-tool count and recovery rate. These differences describe this model's observed interaction path; they should not be treated as language-level causal effects without reviewed traces and a larger or replicated design.

4. Task and category results

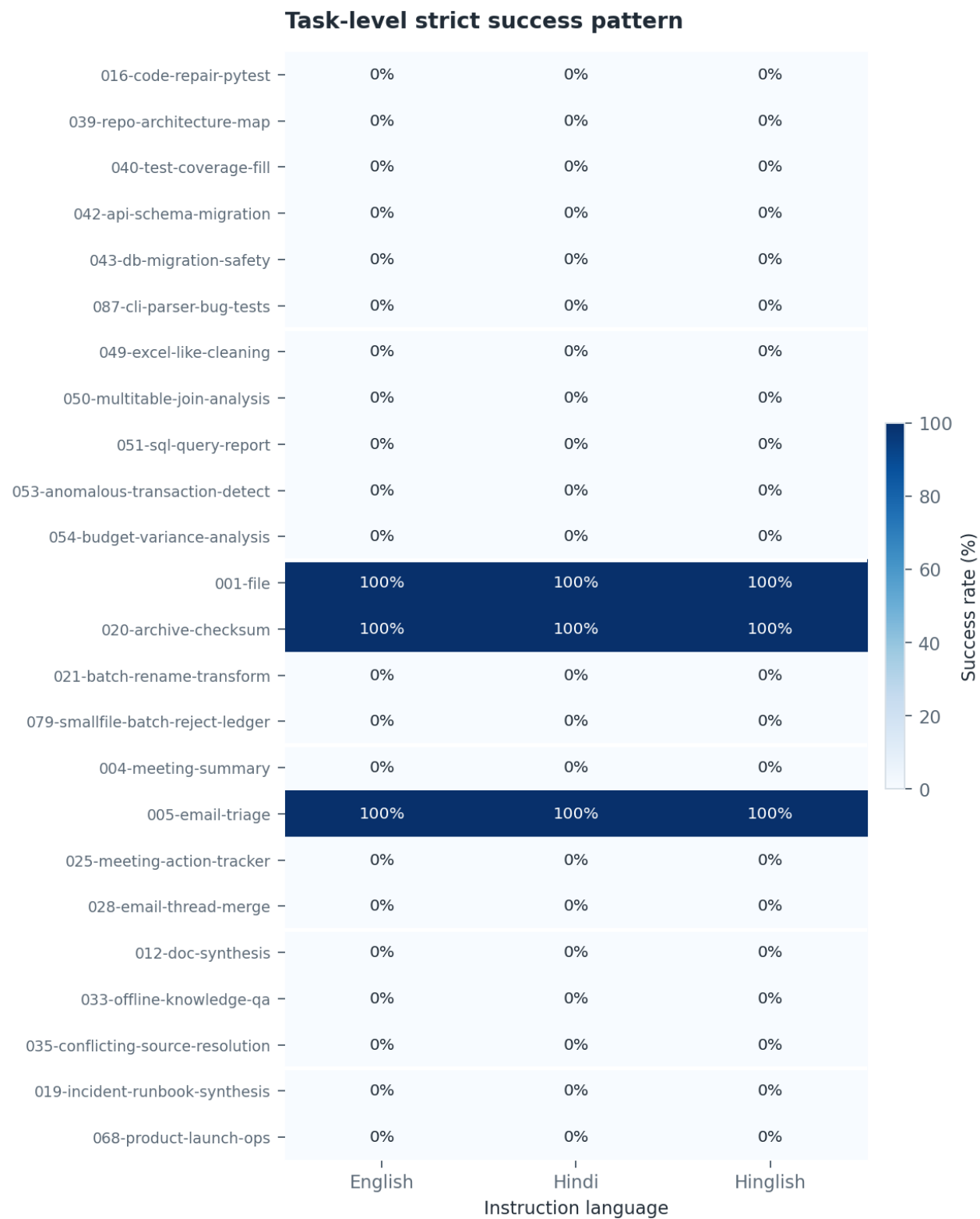


Figure 3. Task-level success pattern. Each cell is the proportion of three repetitions that reached the strict threshold. White separators mark category changes.

Category summary

Category	Tasks	English	Hindi	Hinglish
Software engineering	6	0.0%	0.0%	0.0%
Data and file analysis	5	0.0%	0.0%	0.0%
Workspace and shell	4	50.0%	50.0%	50.0%
Office workflows	4	25.0%	25.0%	25.0%
Evidence and retrieval	3	0.0%	0.0%	0.0%
Multistep workflows	2	0.0%	0.0%	0.0%

Task-level appendix

Task	Category	EN	HI	Hinglish	Oracle score range
016-code-repair-pytest	Software engineering	0.0%	0.0%	0.0%	0.60-0.70
039-repo-architecture-map	Software engineering	0.0%	0.0%	0.0%	0.00-0.80
040-test-coverage-fill	Software engineering	0.0%	0.0%	0.0%	0.10-0.54
042-api-schema-migration	Software engineering	0.0%	0.0%	0.0%	0.00-0.00
043-db-migration-safety	Software engineering	0.0%	0.0%	0.0%	0.06-0.92
087-cli-parser-bug-tests	Software engineering	0.0%	0.0%	0.0%	0.14-0.45
049-excel-like-cleaning	Data and file analysis	0.0%	0.0%	0.0%	0.13-0.49
050-multitable-join-analysis	Data and file analysis	0.0%	0.0%	0.0%	0.10-0.53
051-sql-query-report	Data and file analysis	0.0%	0.0%	0.0%	0.04-0.95
053-anomalous-transaction-detect	Data and file analysis	0.0%	0.0%	0.0%	0.69-0.96
054-budget-variance-analysis	Data and file analysis	0.0%	0.0%	0.0%	0.43-0.69
001-file	Workspace and shell	100.0%	100.0%	100.0%	1.00-1.00
020-archive-checksum	Workspace and shell	100.0%	100.0%	100.0%	1.00-1.00
021-batch-rename-transform	Workspace and shell	0.0%	0.0%	0.0%	0.00-0.18
079-smallfile-batch-reject-leader	Workspace and shell	0.0%	0.0%	0.0%	-
004-meeting-summary	Office workflows	0.0%	0.0%	0.0%	0.00-0.89
005-email-triage	Office workflows	100.0%	100.0%	100.0%	1.00-1.00
025-meeting-action-tracker	Office workflows	0.0%	0.0%	0.0%	0.06-0.83
028-email-thread-merge	Office workflows	0.0%	0.0%	0.0%	0.09-0.82
012-doc-synthesis	Evidence and retrieval	0.0%	0.0%	0.0%	0.40-0.75
033-offline-knowledge-qa	Evidence and retrieval	0.0%	0.0%	0.0%	0.63-0.97
035-conflicting-source-resolution	Evidence and retrieval	0.0%	0.0%	0.0%	0.00-0.81
019-incident-runbook-synthesis	Multistep workflows	0.0%	0.0%	0.0%	0.81-0.85
068-product-launch-ops	Multistep workflows	0.0%	0.0%	0.0%	0.00-0.92

Oracle score ranges are shown only where the upstream oracle returned an explicit numeric outcome score; judge-only tasks are shown as '-'.

5. Limitations and research interpretation

Limitation	Implication
Translation status	Hindi and Hinglish were authored before execution, but the team has not yet approved them. A translation defect could affect both model behavior and the interpretation of language comparisons.
Task sample	This is a fixed 24-task selection from Harness-Bench, not a random sample of all possible office, shell, retrieval, or software tasks. Task-balanced estimates describe this selection.
Model and agent	Only one university-GPU model and one ReAct agent were tested. The result cannot be generalized to other models, agents, serving parameters, or prompting policies.
Repetitions	Three repetitions per cell improve observability but are not a substitute for broad random seeds or independent model snapshots. Temperature was fixed at zero.
Failure attribution	The traces record observable stages, tool results, and recovery behavior. Claims about whether the model, translation, tool, fixture, or oracle caused a failure should wait for human annotation.
Privacy and reproducibility	The report omits the private endpoint, bearer key, and server-backed model identifier. The exact model manifest, raw SQLite database, traces, and workspace artifacts remain local and ignored; the tracked protocol records how to reproduce them with private runtime access.

Recommended next research steps

1. Review every Hindi and Hinglish overlay against the protected-entity checklist; freeze an approved dataset version without rewriting historical runs.
2. Annotate a representative sample of traces by failure stage before making causal claims about translation or model behavior.
3. Replicate the language comparison with additional models or agents only after the Phase I protocol and review workflow are stable.

Appendix: reproducibility and artifacts

The tracked implementation is on branch prishiv_dev. The main study was run as experiment phase1_language_comparison. Raw results and private runtime metadata are intentionally ignored by Git; a local run-index.csv maps each stable cell_id to its trace and result JSON.

Artifact	Purpose
docs/phase1_protocol.md	Secret-free study contract, preflight, execution, review, and completion criteria.
benchmark/task_selection.yaml	Pinned source revision, exact 24-task selection, quotas, and source hashes.
benchmark/translations/phase1.yaml	English originals plus Hindi and Hinglish overlays with review metadata.
data/phase1/experiment/provisional_report.json	Machine-readable findings generated from the main SQLite database; local and ignored.
data/phase1/experiment/run-index.csv	Secret-free local index for all 216 cell logs; local and ignored.
data/phase1/experiment/traces/*.jsonl	Per-cell event logs including tool arguments/results and usage metadata; local and ignored.

This report was generated from the completed database after the final preflight and test audit. Findings should be cited as provisional until translation review and publication decisions are complete.